Aditya Verma
Purdue EE 2028

A Collection of Cool Things I've Made

Brushless DC Motor + Controller

Designed 36-slot, 42-pole **8010 outrunner BLDC**, 3D printed housing

Hand-wound 3 phases (wye terminated) with 6 parallel, 26 gauge enameled wire for high torque-per-amp, minimal cogging, low-kV

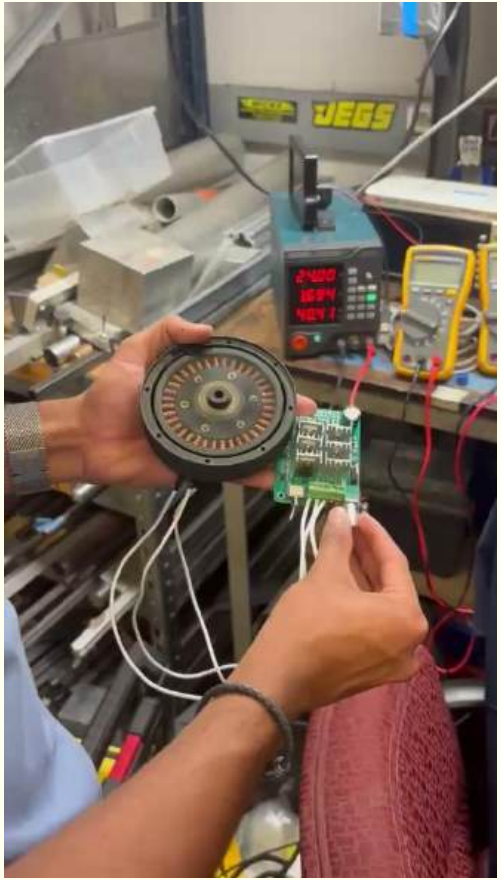
Building **FOC-capable controller PCB**: 10 A/phase, 2 oz copper, Kelvin-sensed shunts, independent on/off gate edge tuning

STMG0 integrated with gate driver, SPI for **hall effect** (positional feedback based on magnet), **UART to USB bridge** for PC connection

Tuned system for **2-97% duty cycle at 20 kHz switching,** optimizing current ripple, switching losses, and bootstrap cap sizing

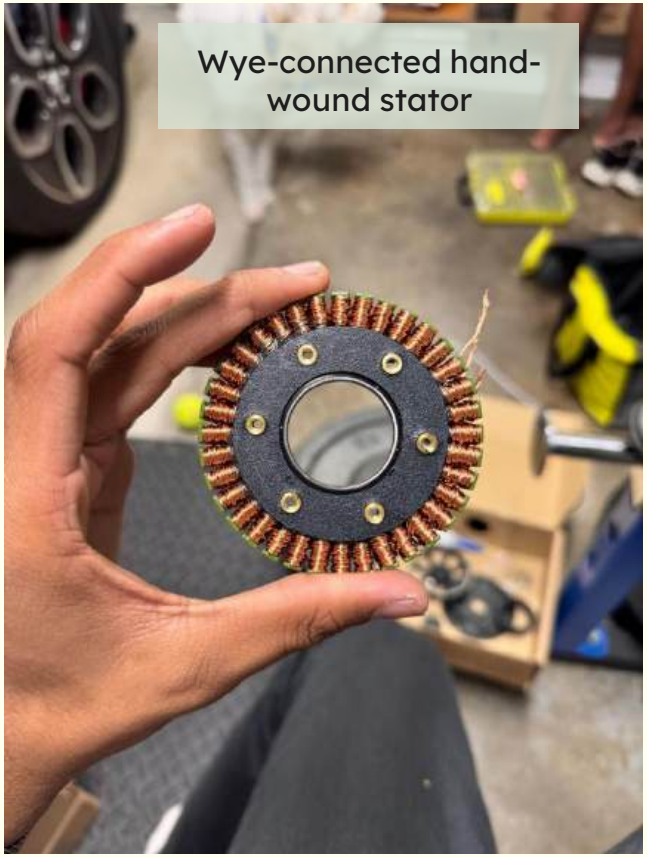
Integrated **reverse-polarity** and **over-voltage** protection with regenerative braking capability for 40 V LiPo operation

4-layer PCB layout focused on high current capability, **analog signal integrity** (Kelvin-sensed shunts), and **thermal dissipation**

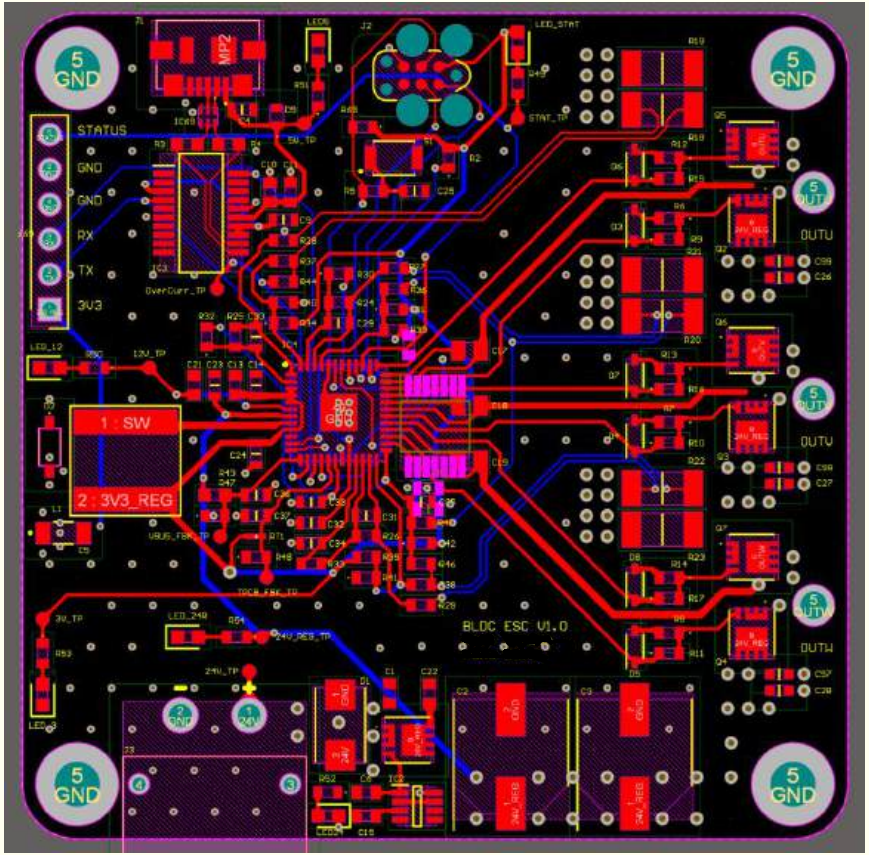


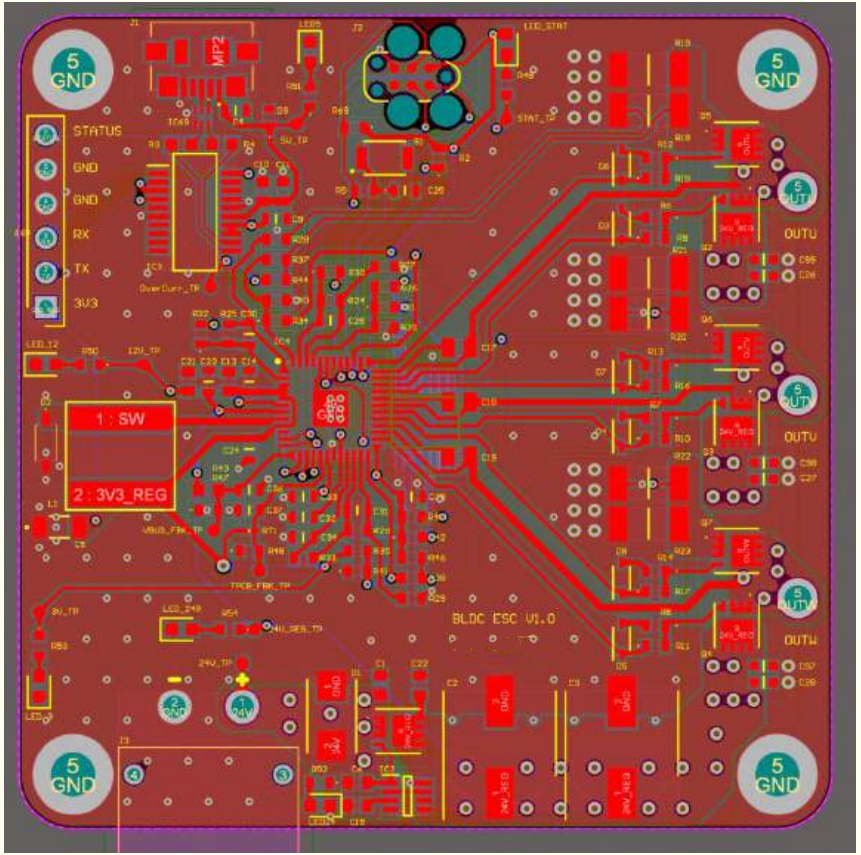
Brushless DC Motor + Controller



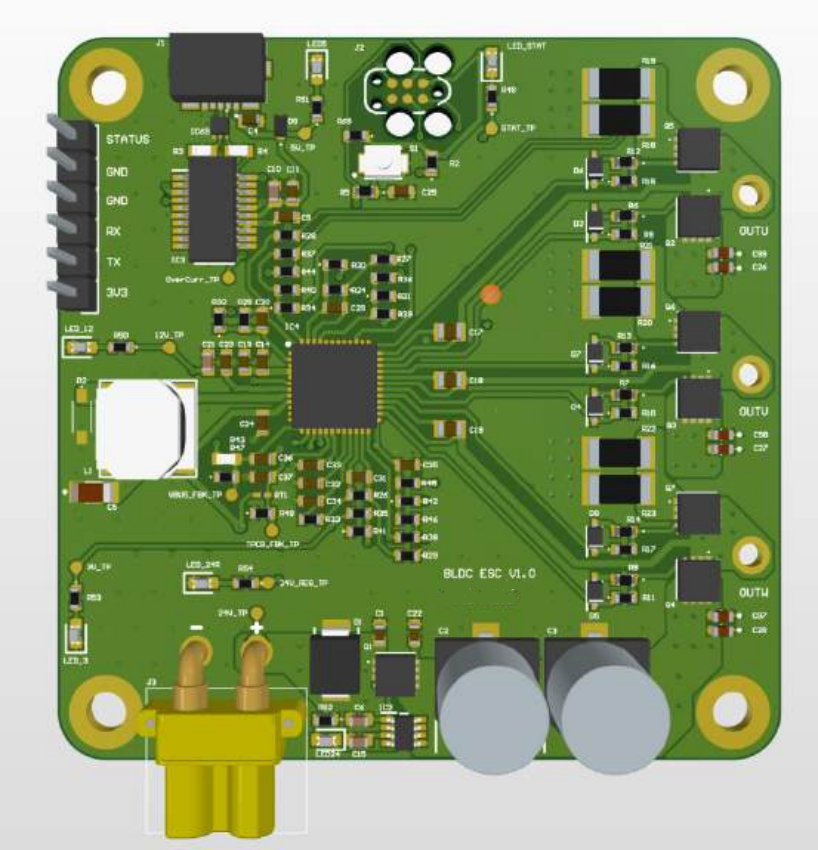


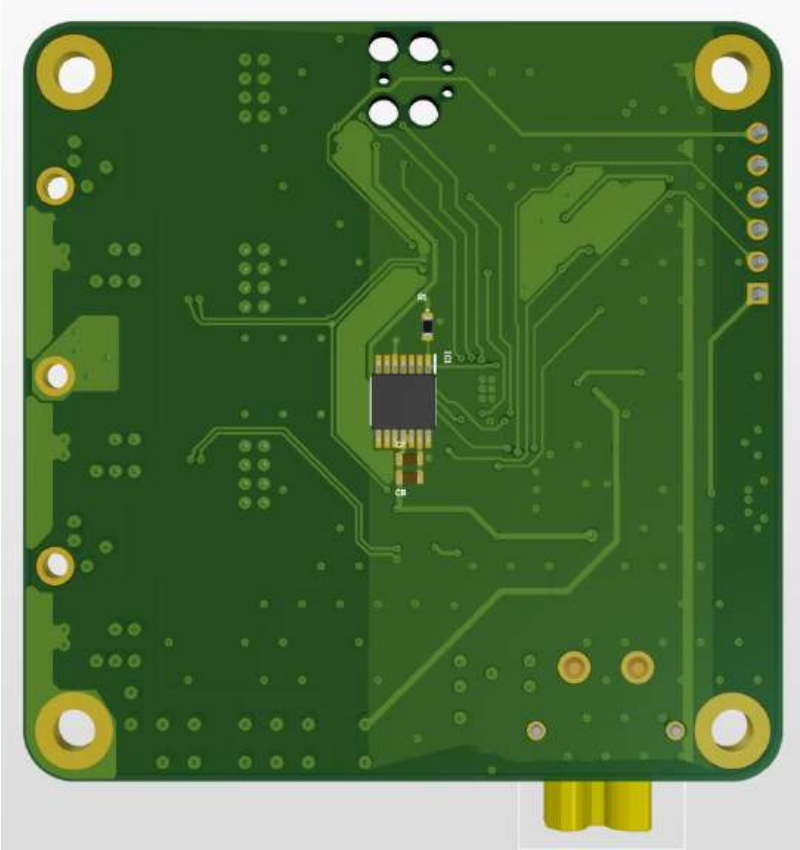
BLDC Controller Layout





BLDC Controller Layout

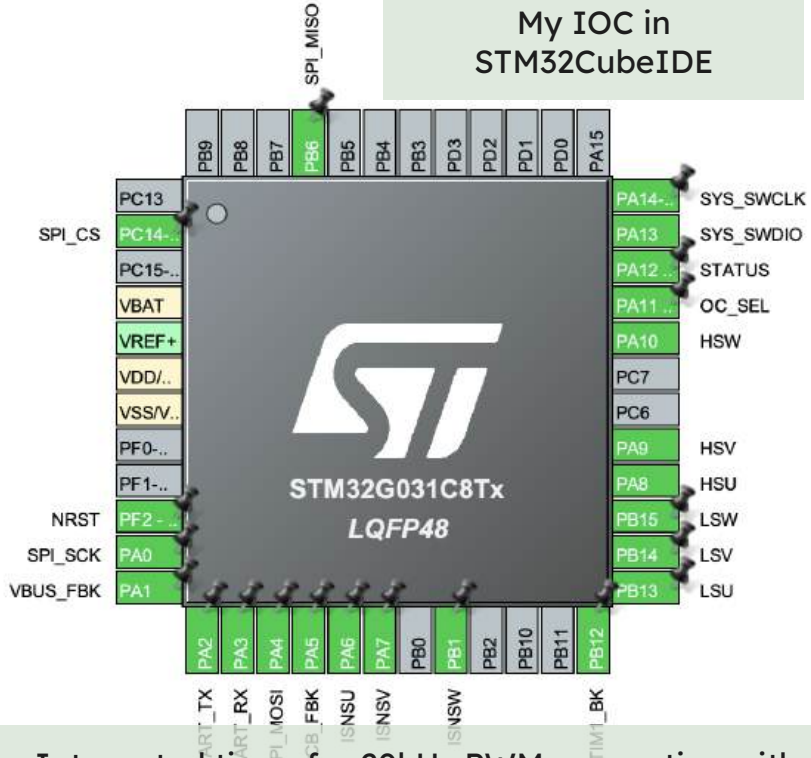




BLDC Controller Soldered + Testing

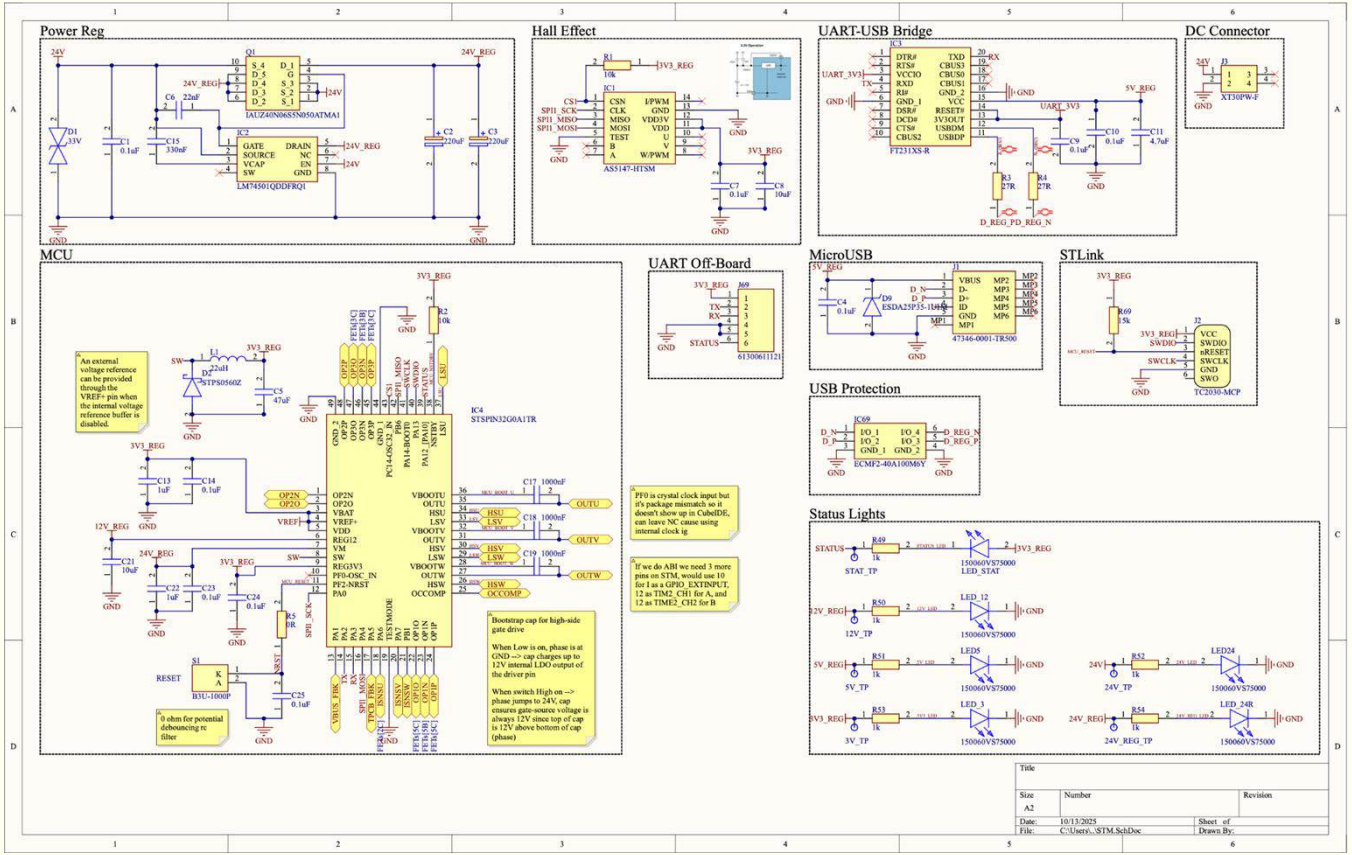


My IOC in STM32CubeIDE

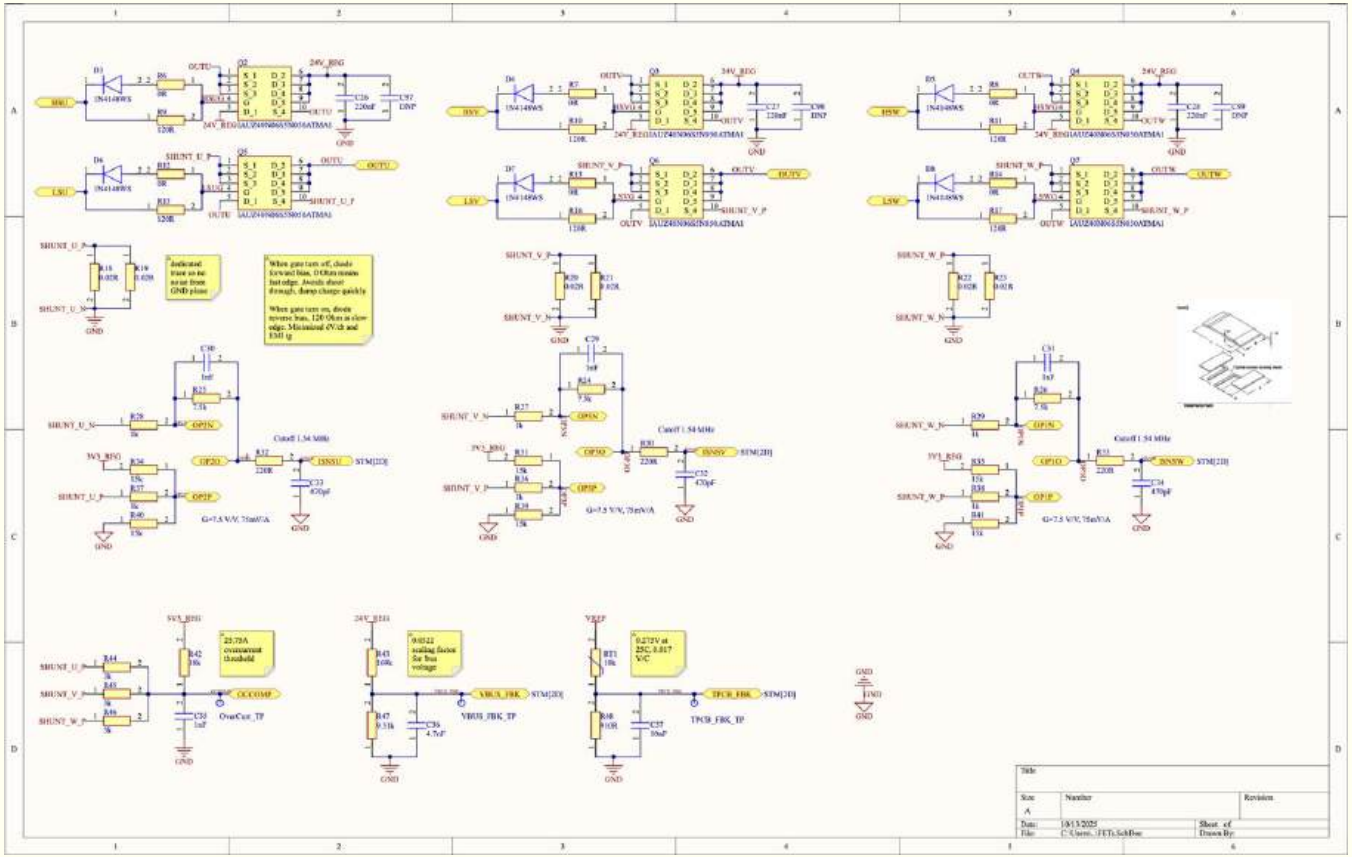


Integrated timer for 20kHz PWM generation with ADC sampling in middle of 50us PWM cycles

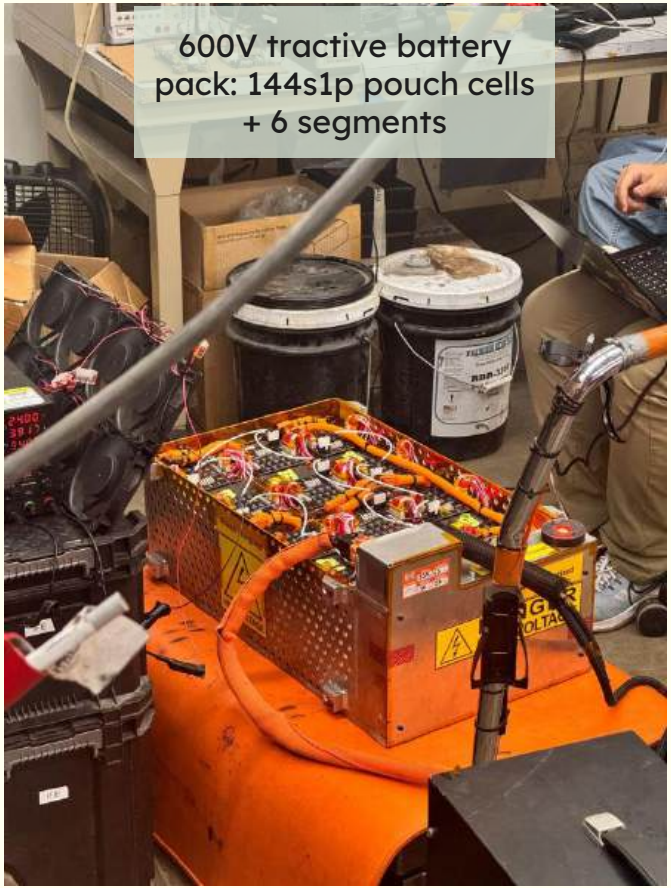
BLDC Schematics

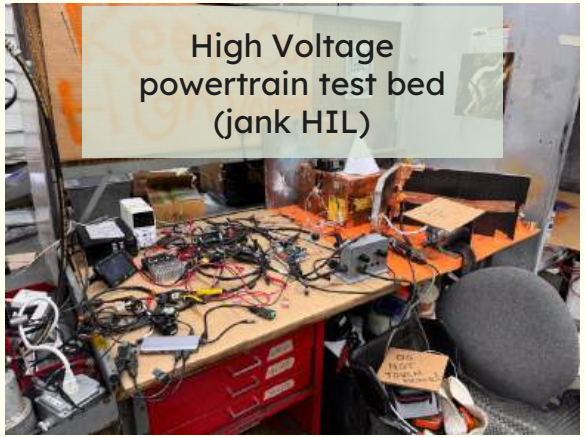


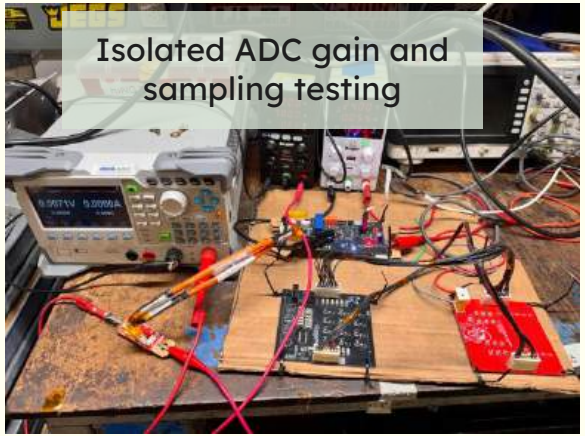
BLDC Schematics



PSR Battery Management System







Lead architecture, design and validation of **600 V battery management system (BMS)** and **high-voltage power electronics**

Manufactured, validated, debugged custom PCBs: Safety (vehicle state machine, SPI, CAN), steering wheel (I^2C)

Optimized and tuned performance of **analog filters and peripherals** with oscilloscope, waveform gen, load tester, and LTspice

BMS Primary PCB

Manages **fault detection, precharge, isolated ADC**, communication with vehicle control board

CAN 1 (car-critical) and CAN 2 (logging) with transceivers to STM

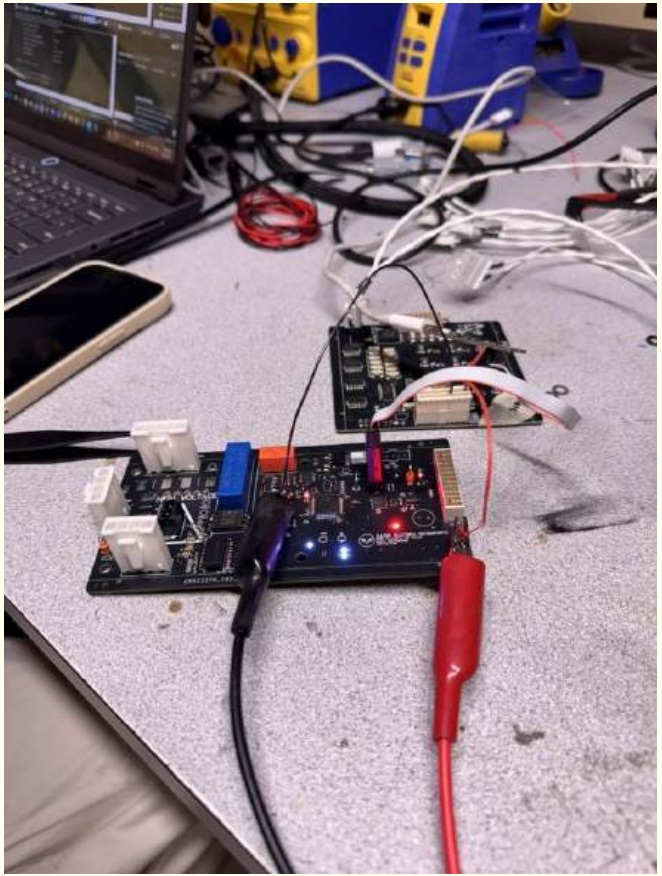
SPI for isoADC to **measure pack status** (for SOC, SOH, SOP)

SPI for flash storage (data logging)

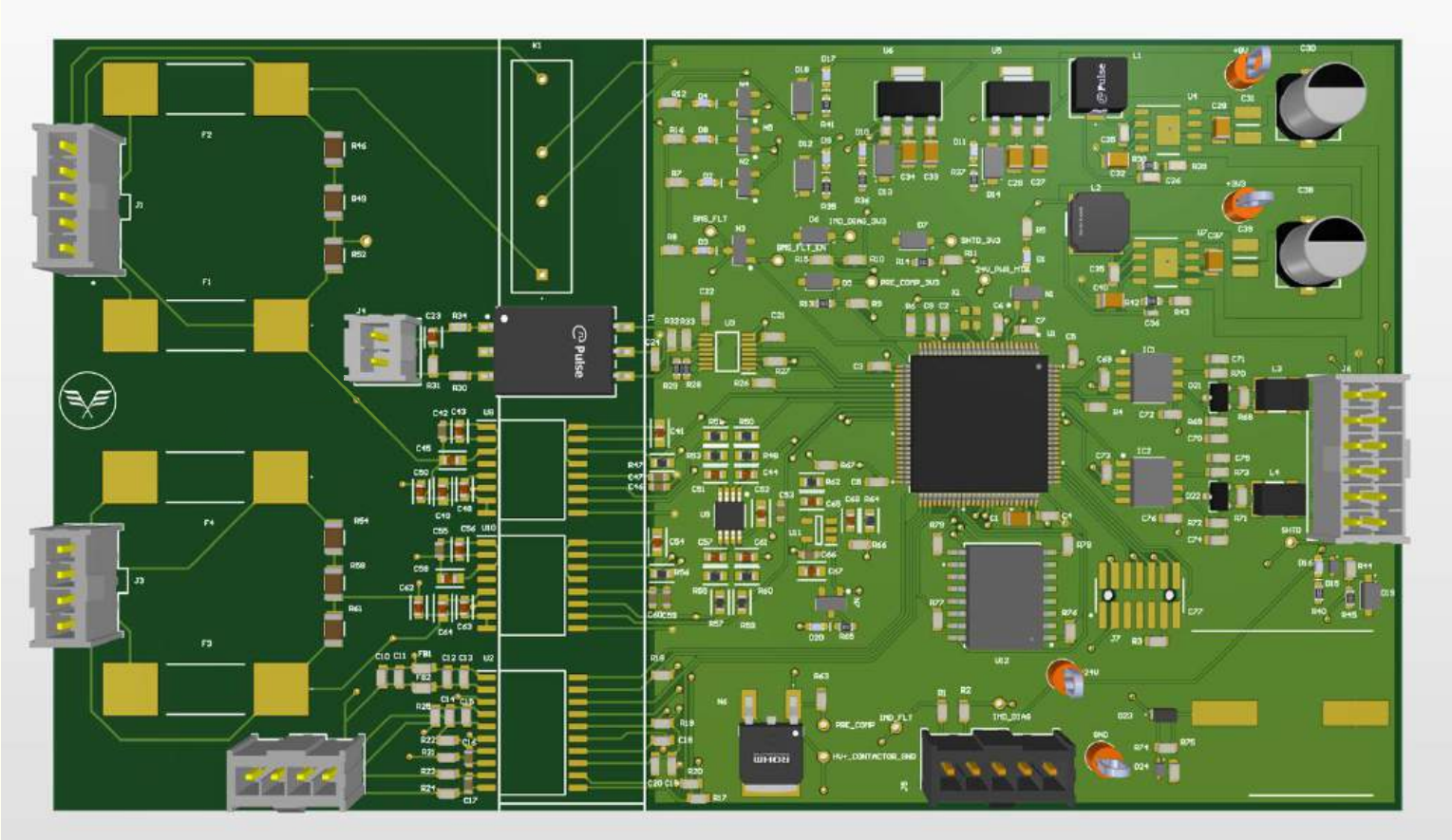
SPI for **communication to secondaries** (isolated through AD proprietary isoSPI with external transformer)

24V - 8V Buck, 8V - 5V LDO, 8V - 3V3 LDO

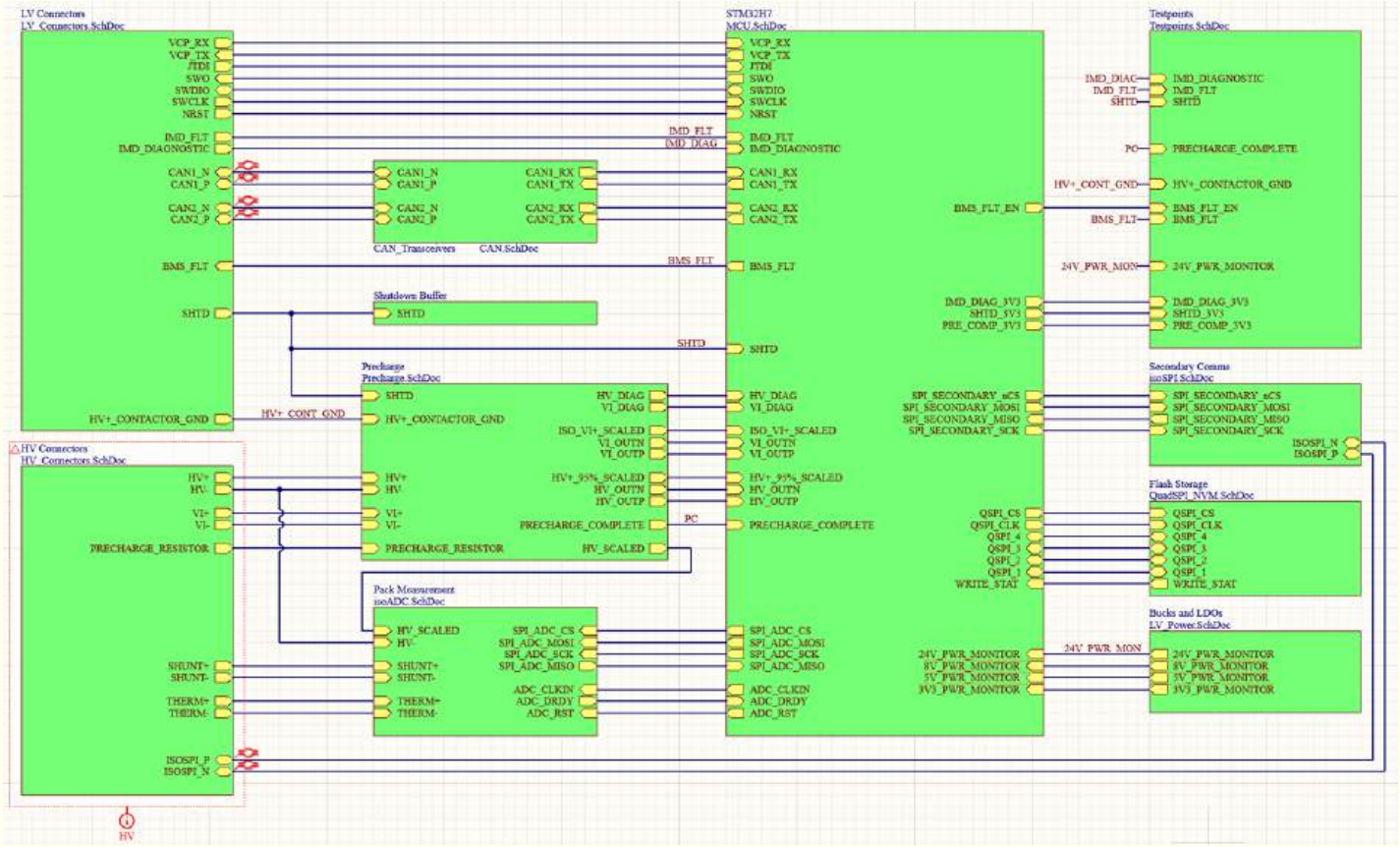
Layout for **HV isolation**, creepage and clearance



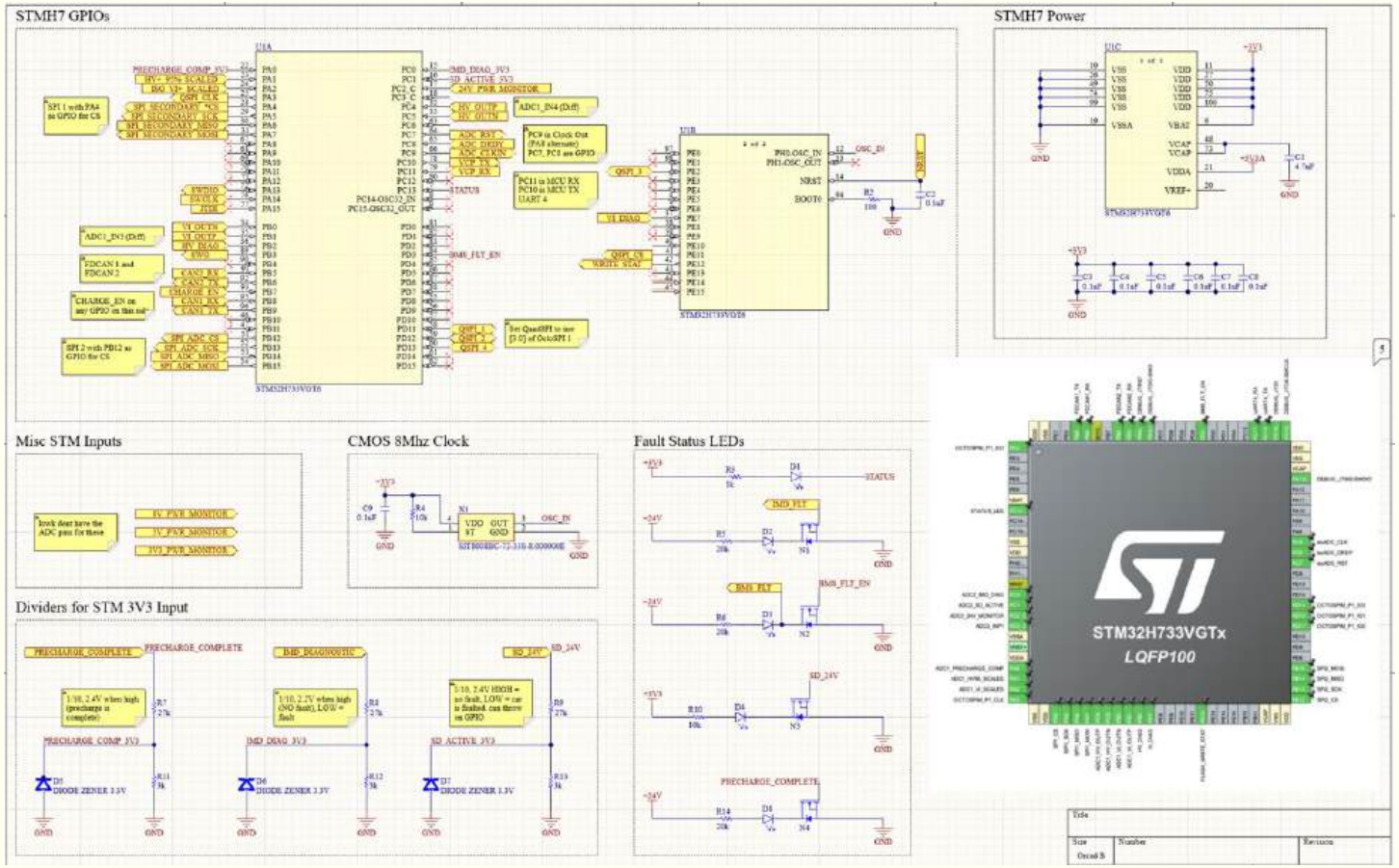
BMS Primary PCB - Layout



BMS Primary PCB - Schematics

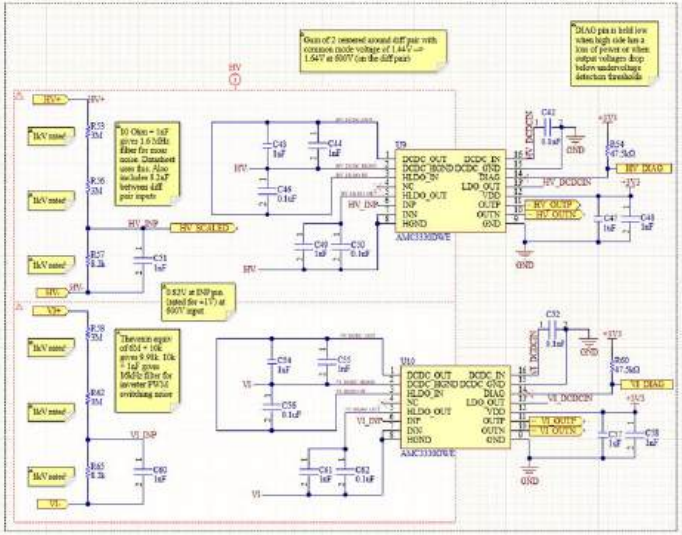


BMS Primary PCB - Schematics

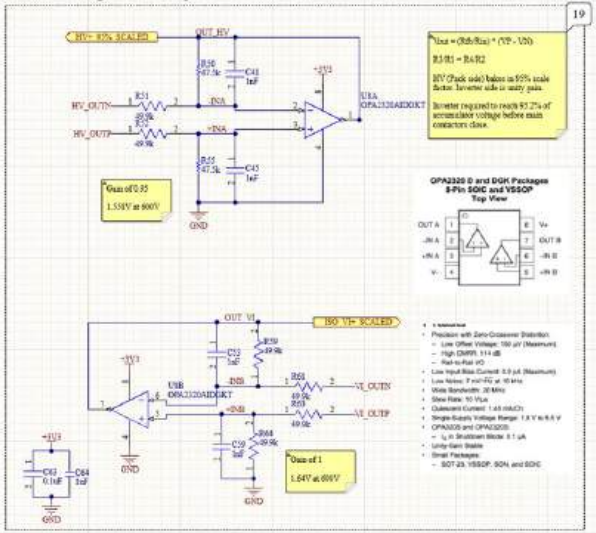


BMS Primary PCB - Schematics

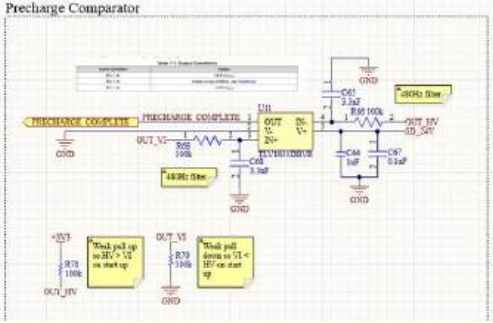
HV 1% Scaling Dividers + Isolation



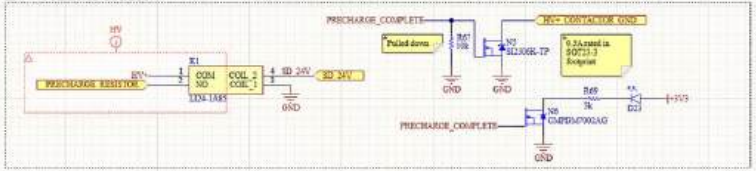
Diff Pair to Single-Ended Output



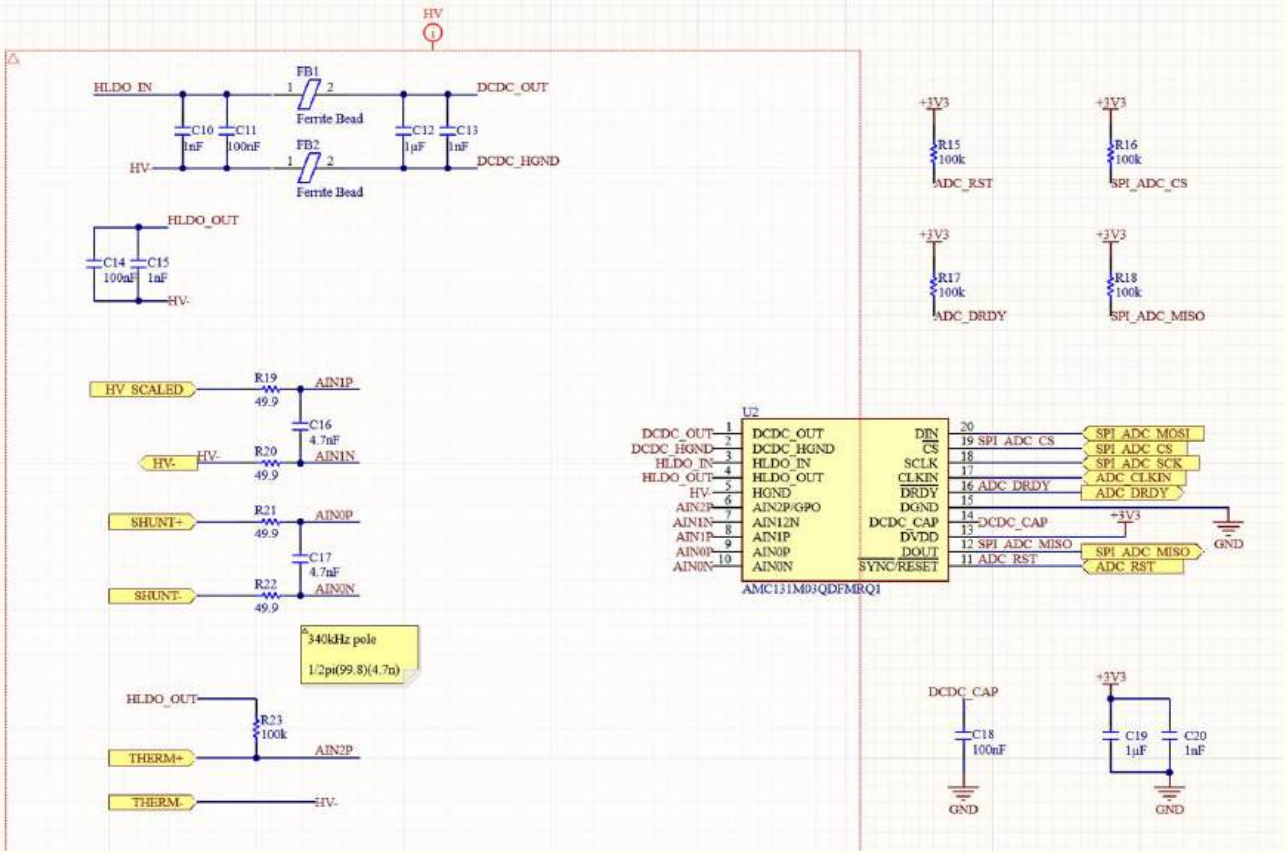
Precharge Comparator

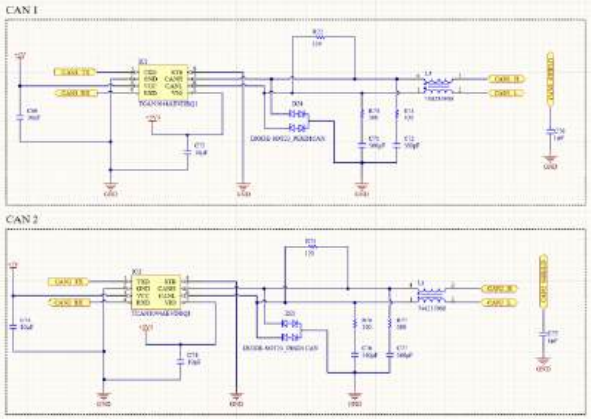


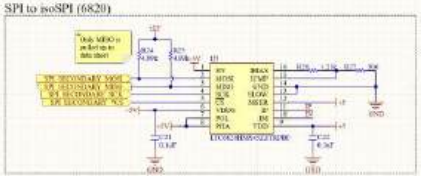
HV+ Contactor Control

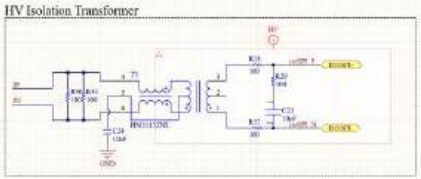


BMS Primary PCB - Schematics

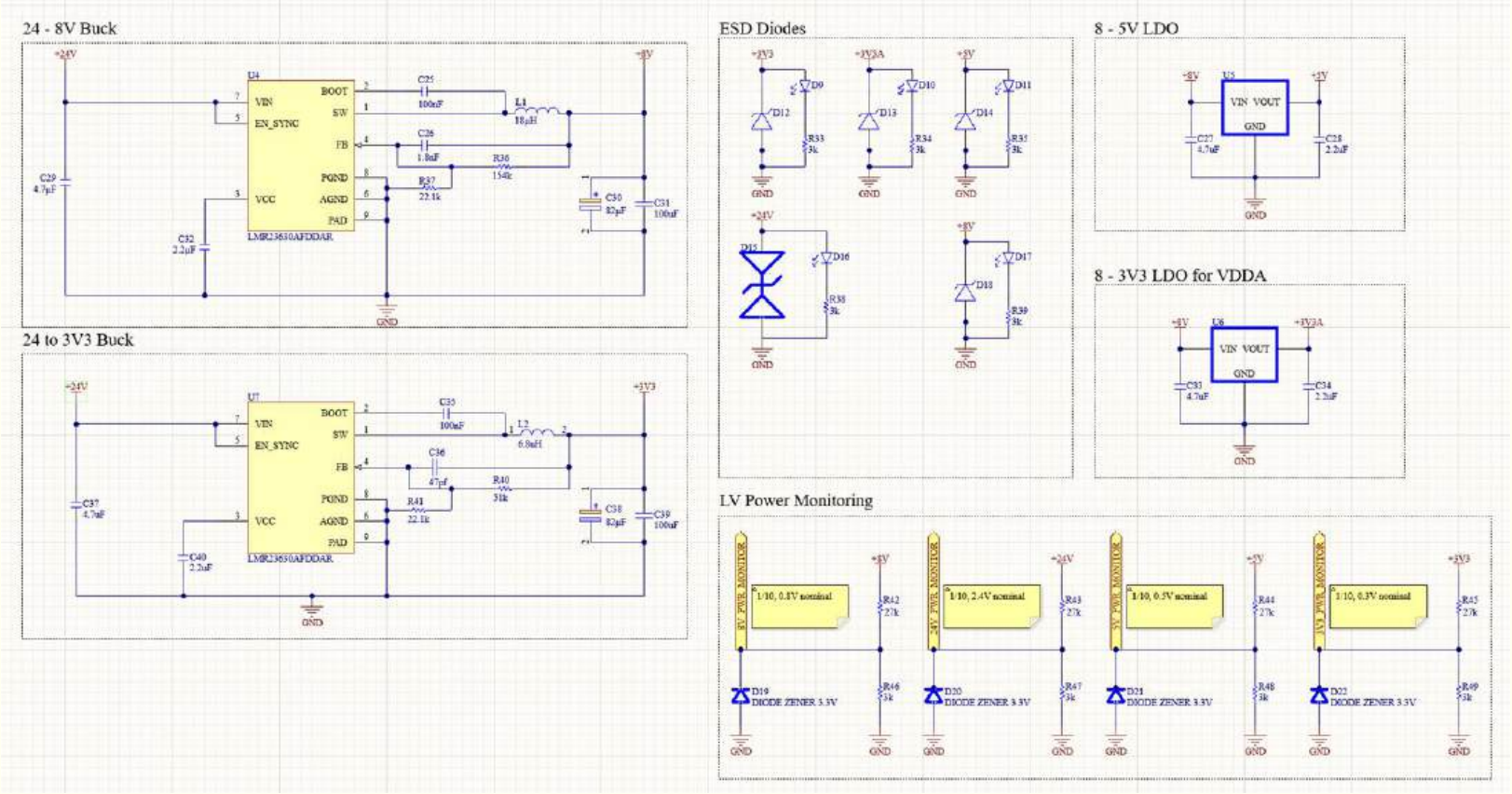






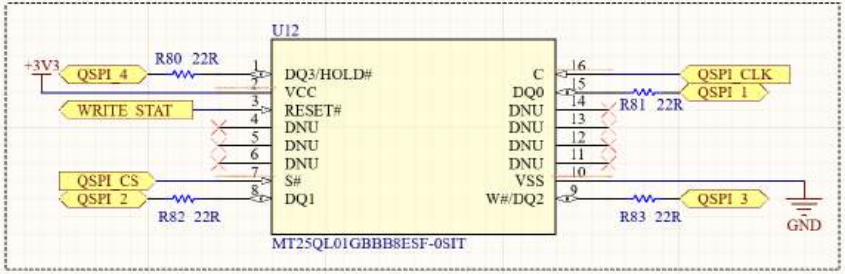


BMS Primary PCB - Schematics

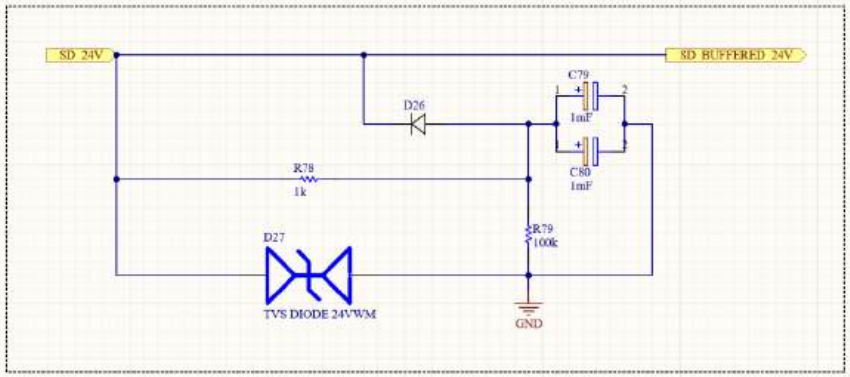


BMS Primary PCB - Schematics

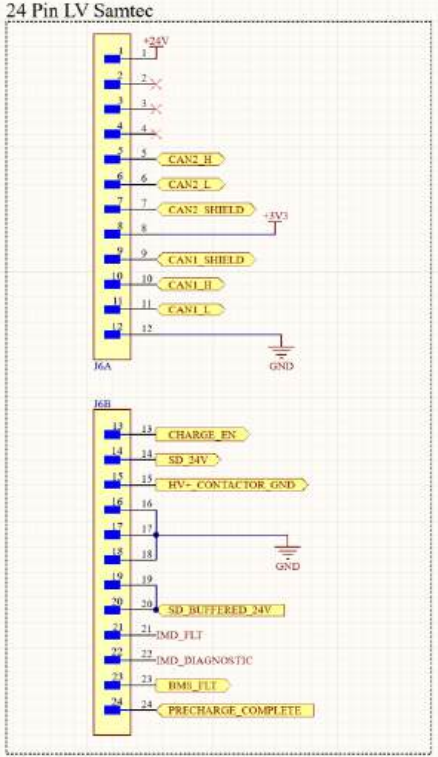
1 Gbit NOR Flash



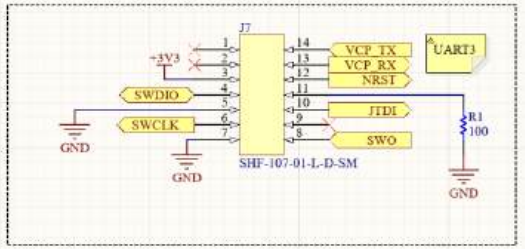
150ms Shutdown Contactor Buffer

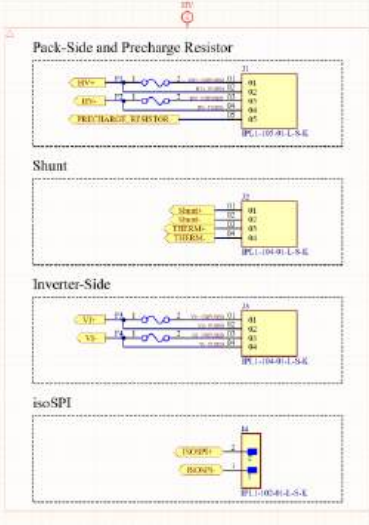


24 Pin LV Samtec



Flashing/Debug





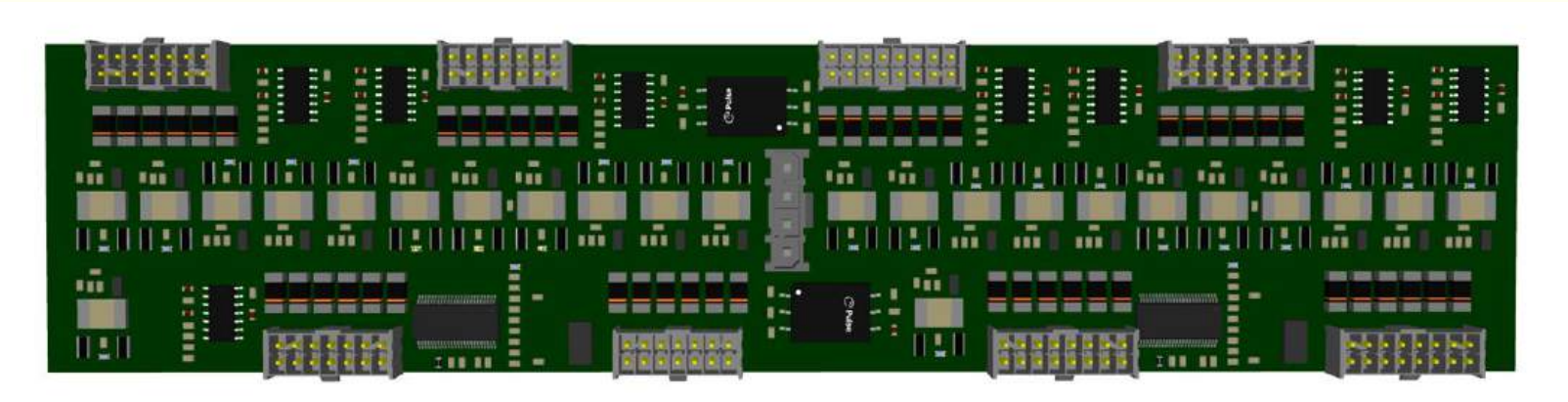
BMS Secondary PCB

Measures **144 individual cell voltages** and temperatures with fused connections
Powered off cells (isolated from LV system)

isoSPI daisy chain for live cell telemetry between 6 secondary PCBs and primary PCB

Op-amp network to measure **hottest-of-3 cells** (limited analog inputs to 6811 IC)

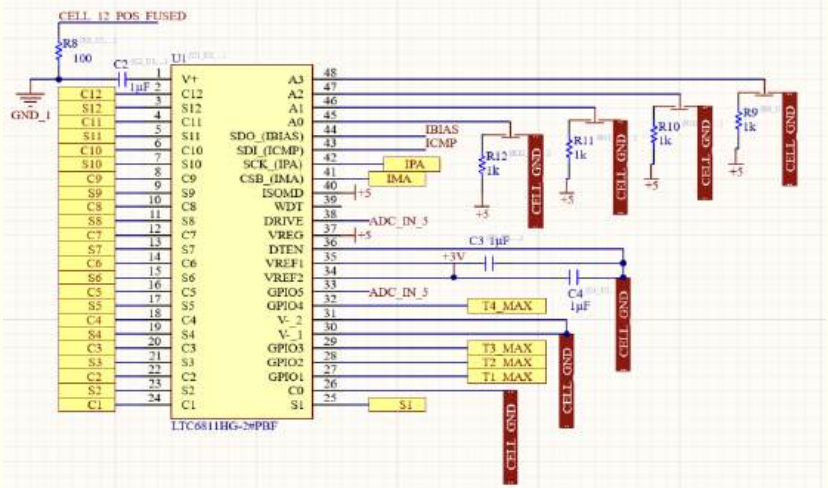
New Design: PCB as segment lid, 2x faster cell balancing with thermal jumpers for dissipation

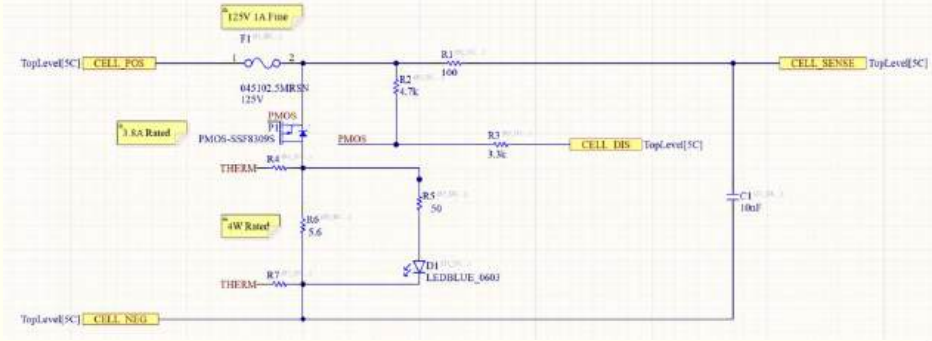


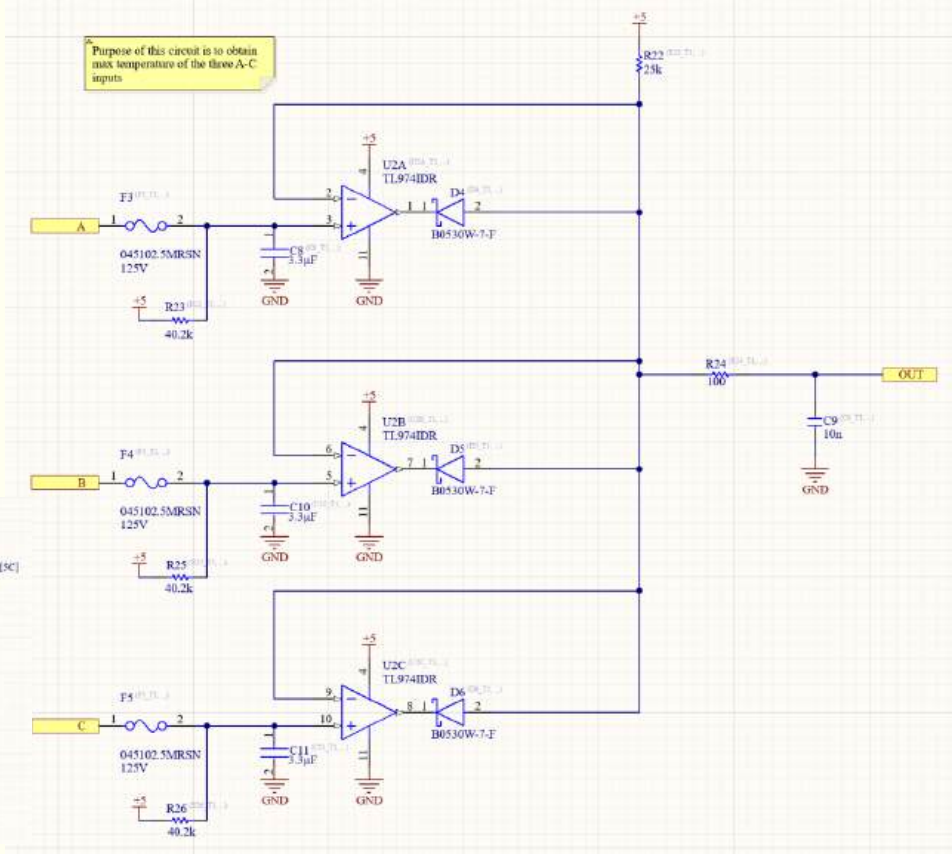
BMS Secondary PCB - Schematics



BMS Secondary PCB - Schematics





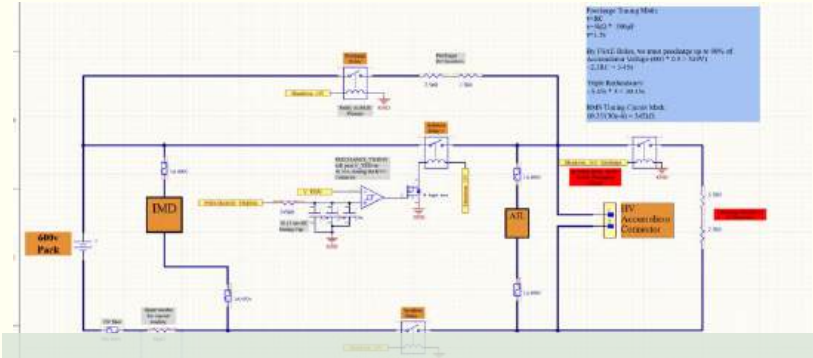


Voltage Based Precharge PCB

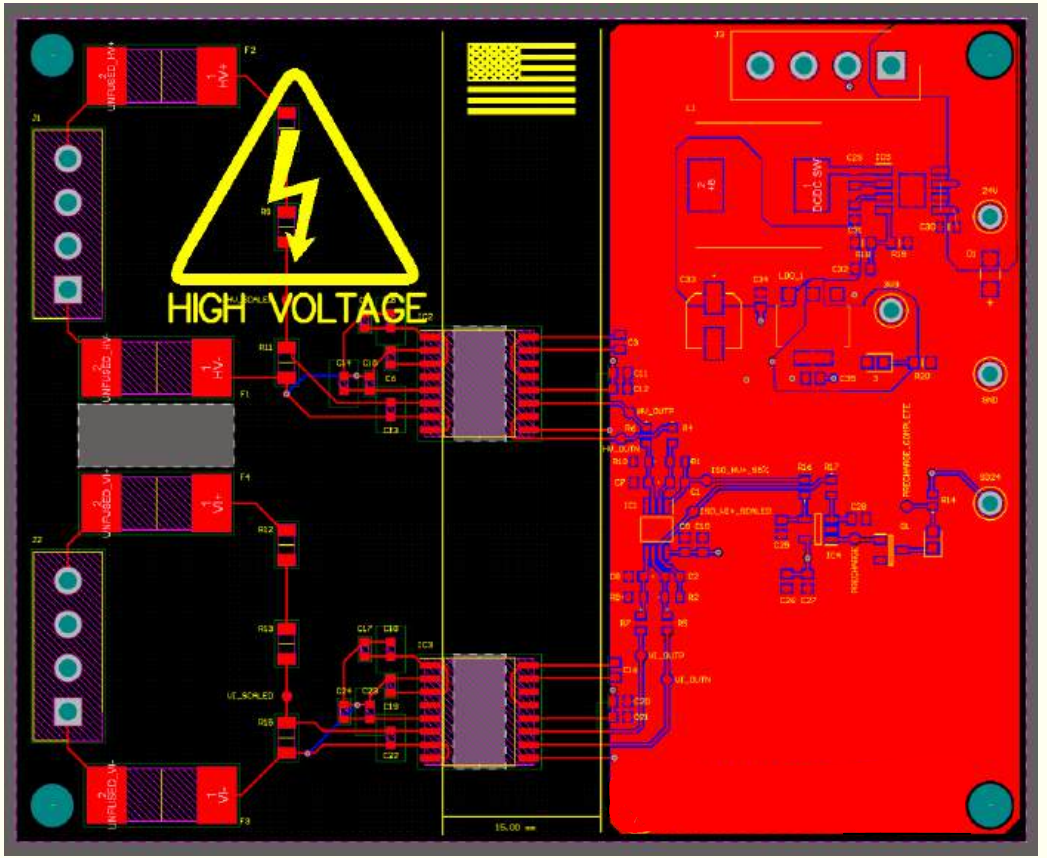
Old: uses large 10s RC timer (~3x more than 3s need for 90% precharge → inverter state is unknown, only assumed

Flaky shutdown line caused precharge failure, blowing main fuse with inrush current

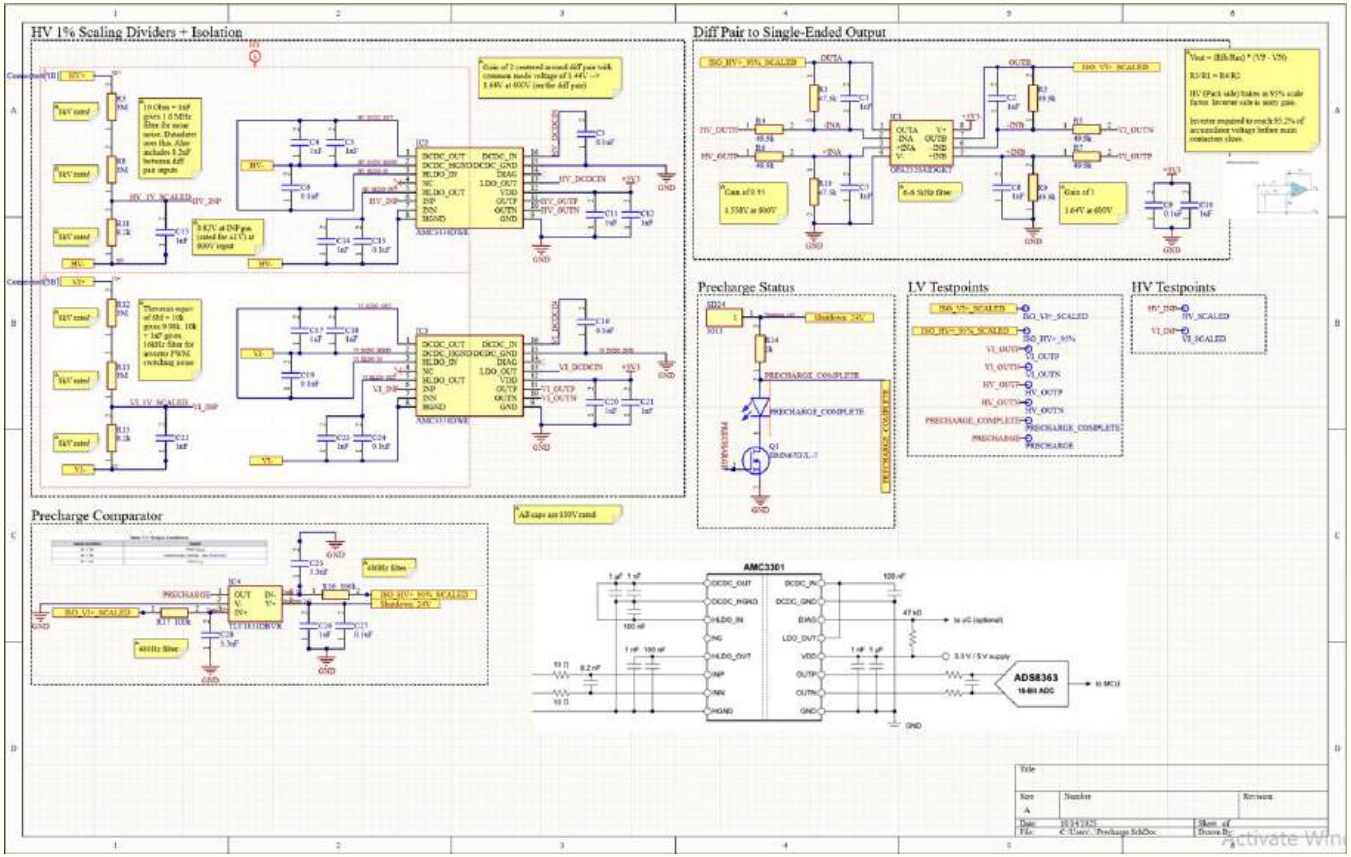
New Design: uses isolated amplifiers to measure and compare pack and inverter voltage directly, only closing contactors once 95% precharge is reached



HV wiring with main and precharge contactors



Voltage Based Precharge PCB



Low Power Dog Collar (Collie)

Designed **1” x 2”, 6-layer PCB** integrating mmWave radar, Wi-Fi, GPS, and IMU, with ultra-low 87 mW power draw for dog behavior tracking

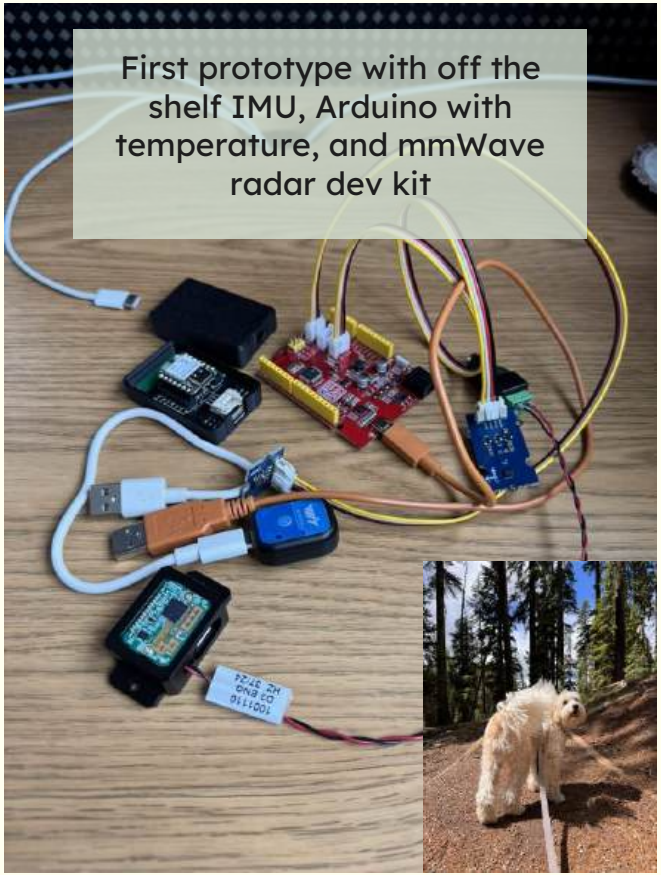
Layout focused on digital/analog layout partitioning, impedance controlled traces, and ground stitching for **mixed-signal integrity**

Built V0 without radar module and **designed V1** with radar

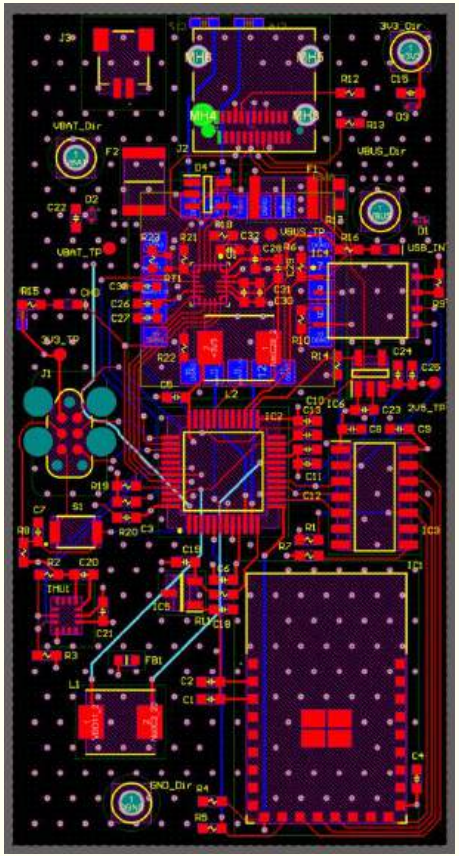
Tuned radar detection thresholds with **TI eval board**, validating 91% algorithm accuracy against PPG ground-truth

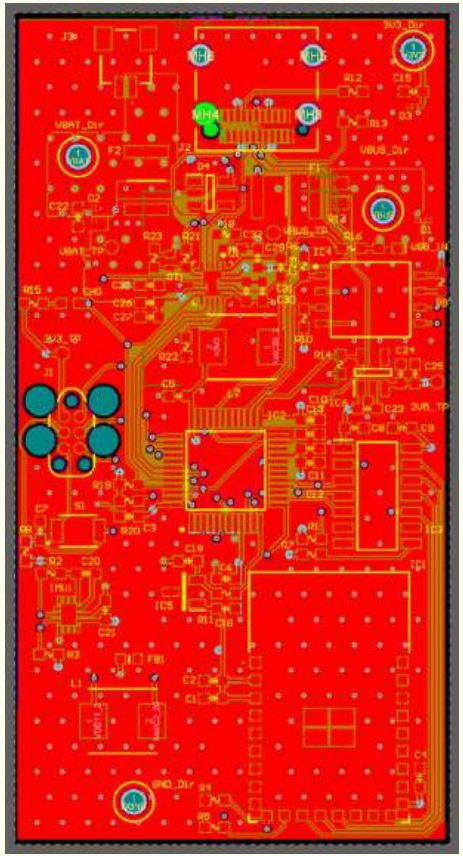
IMU over I2C, TI mmWave radar over UART, USB-C interface integrated with STM32, BMS for LiPo charging and buck regulation, ESD protection

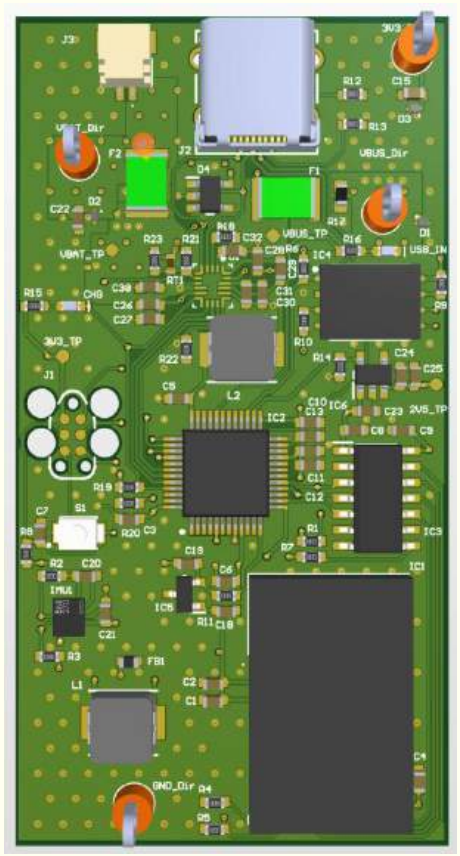
Partnered with Purdue University College of Veterinary Medicine for **planned pilot** and validation program and faculty mentoring

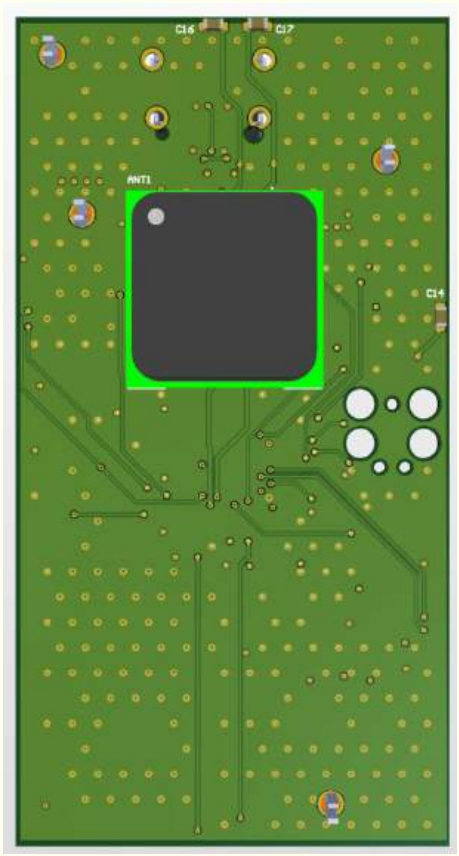


V0 Motion Tracking Collar PCB

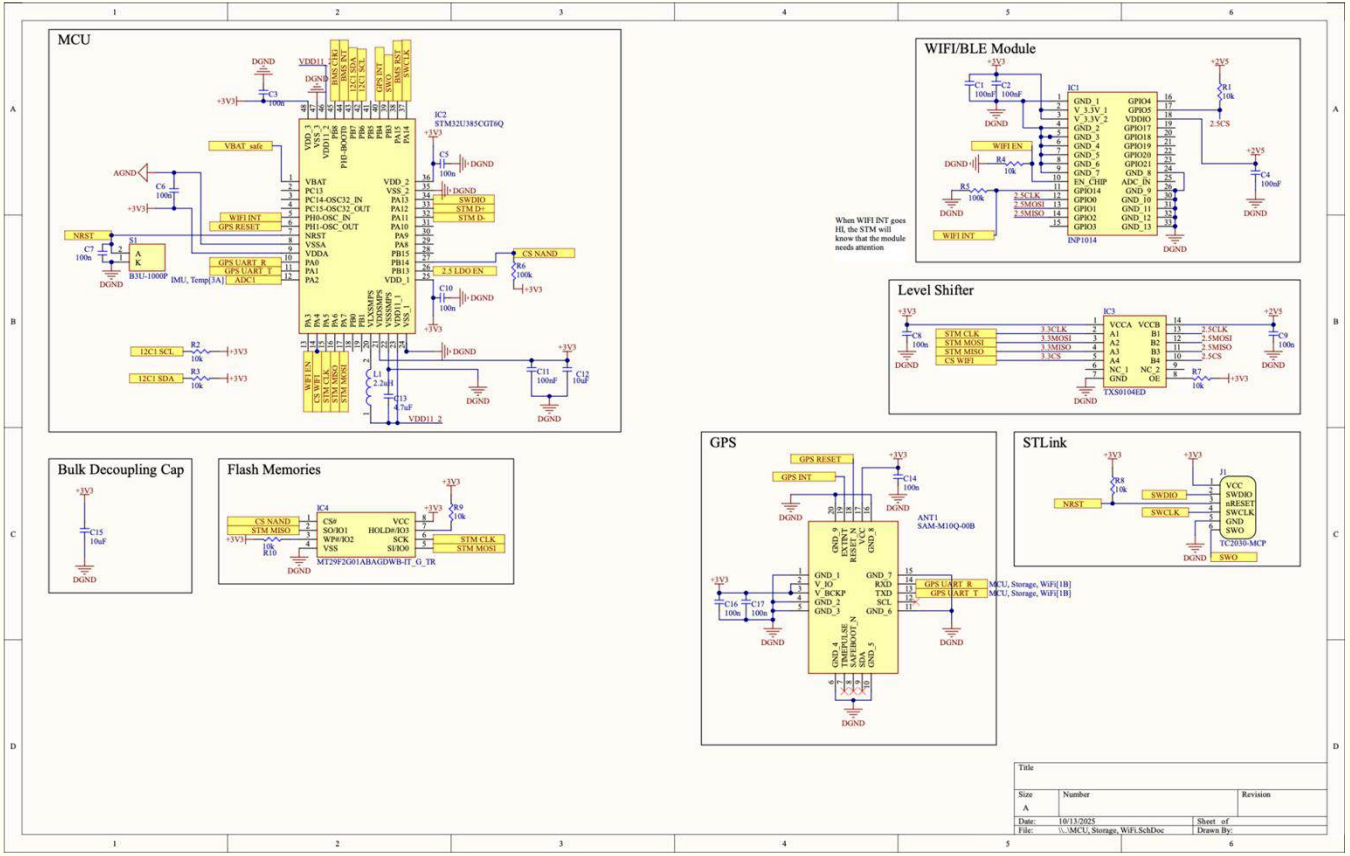




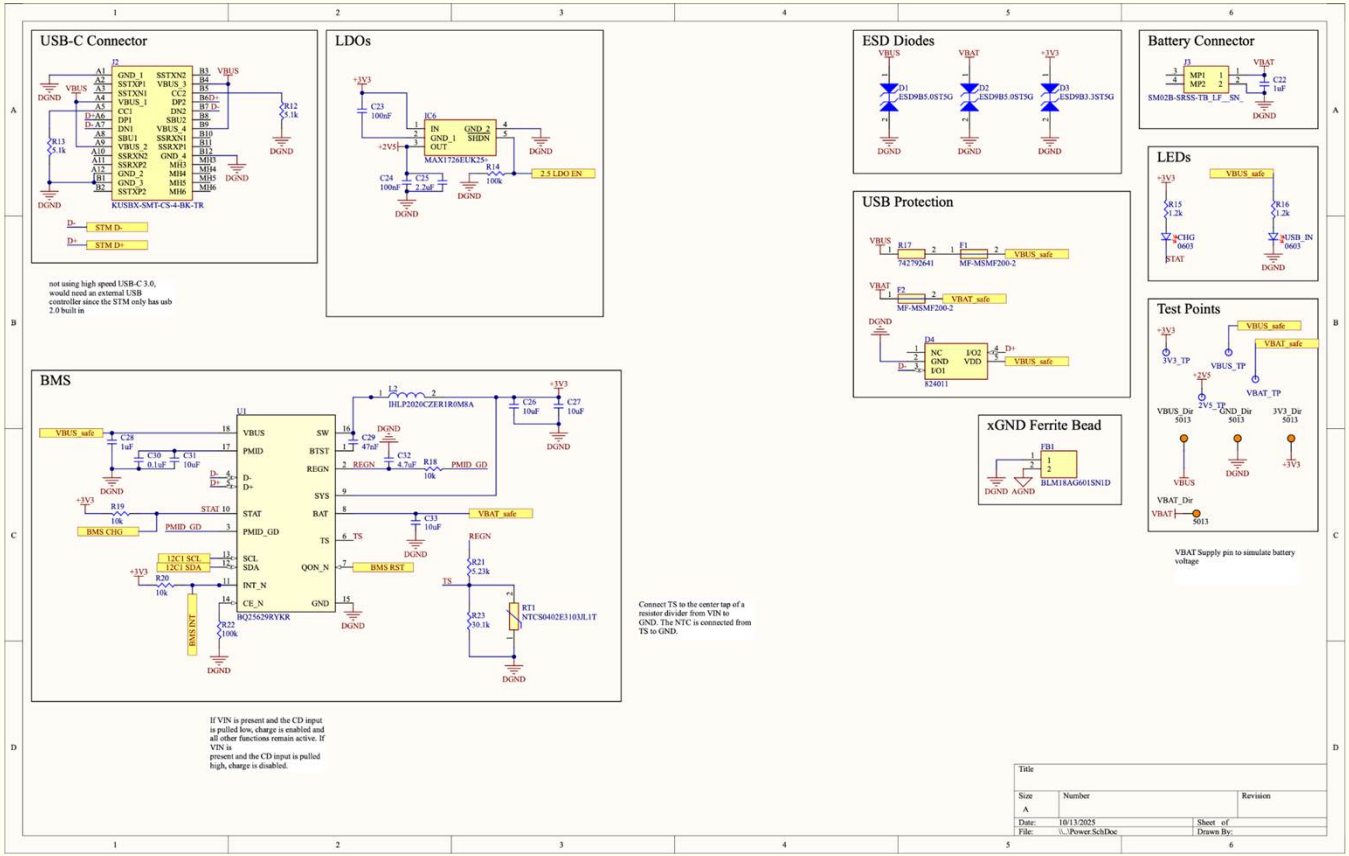




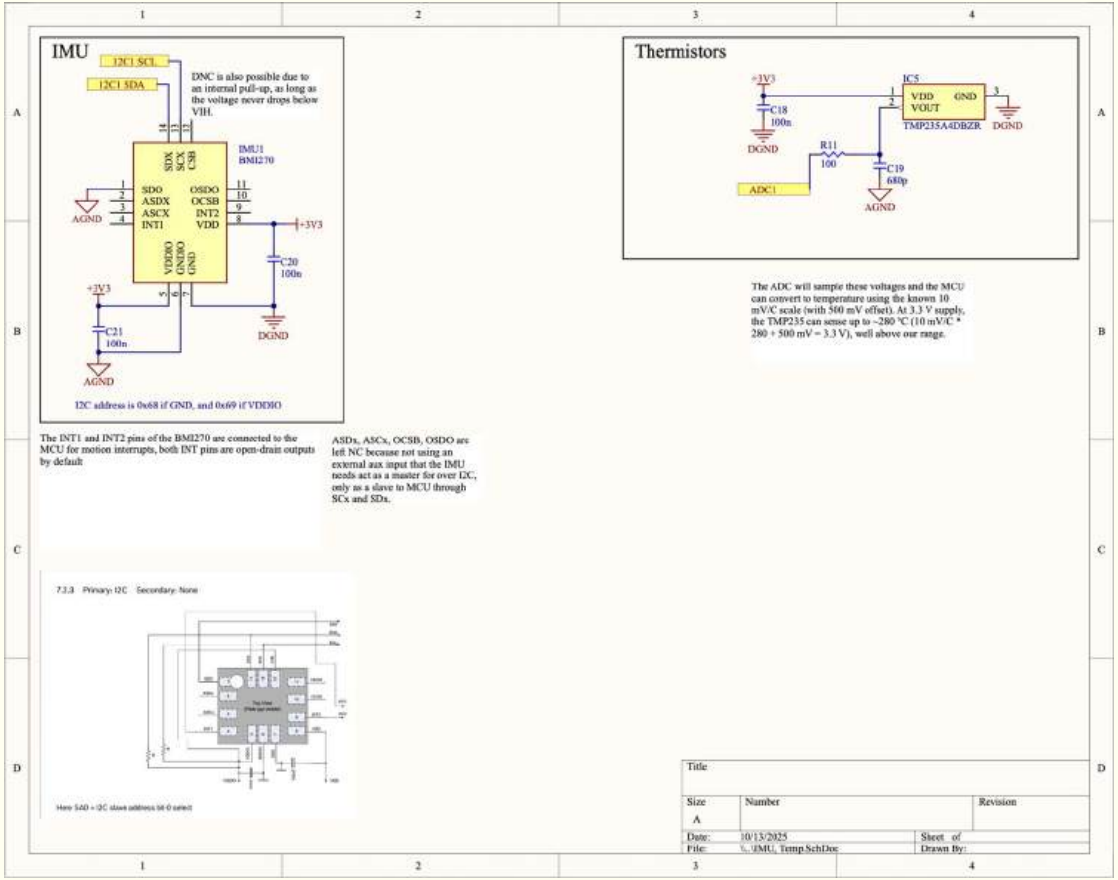
V0 Motion Tracking Collar PCB



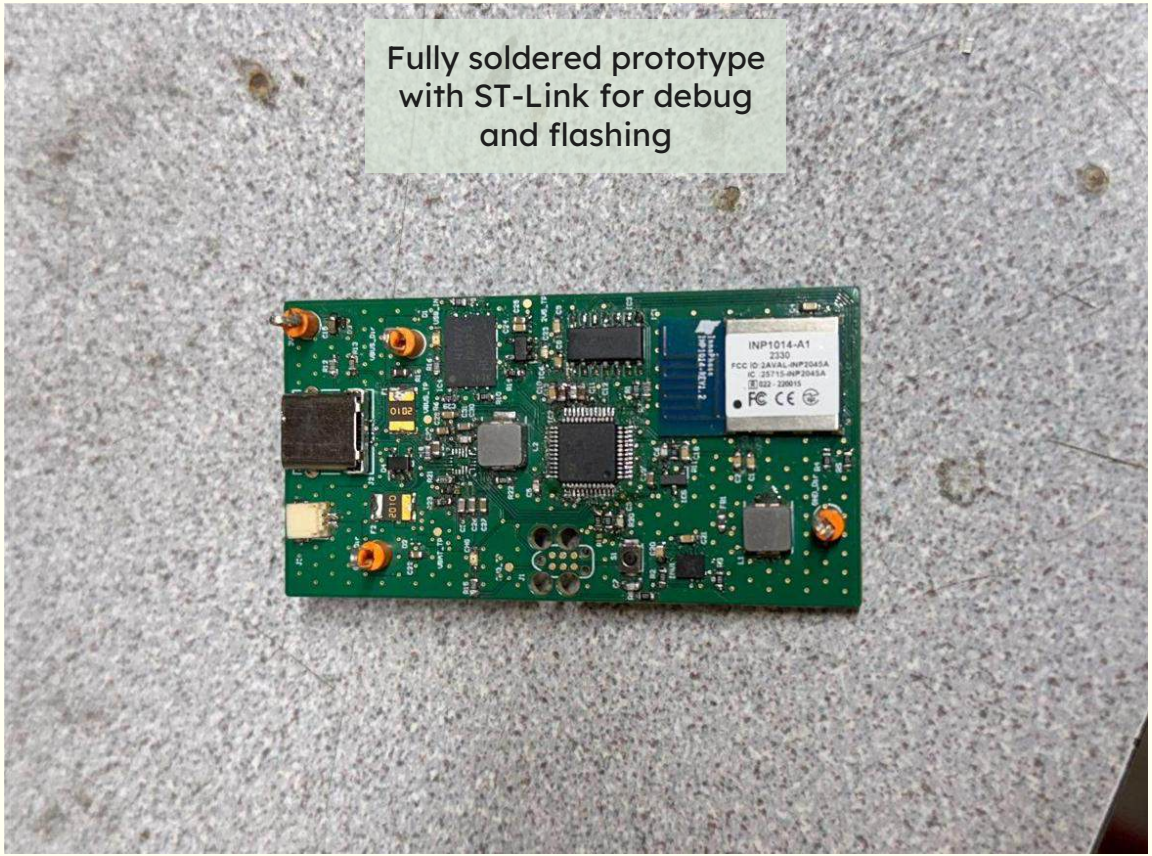
V0 Motion Tracking Collar PCB



V0 Motion Tracking Collar PCB

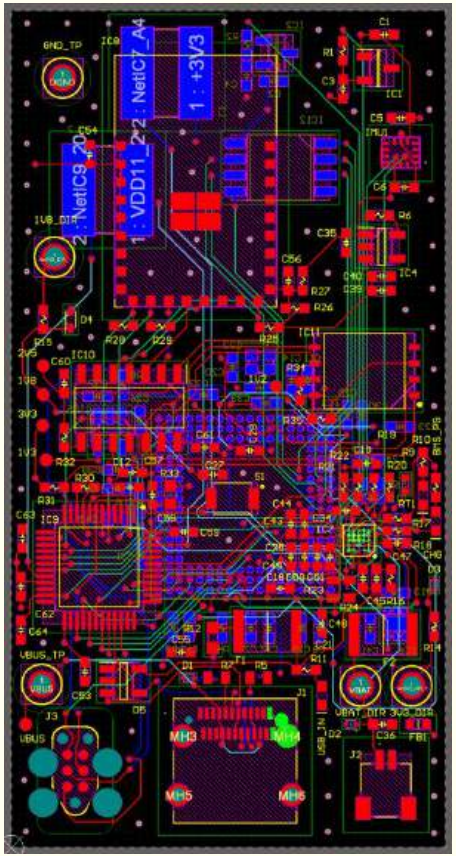


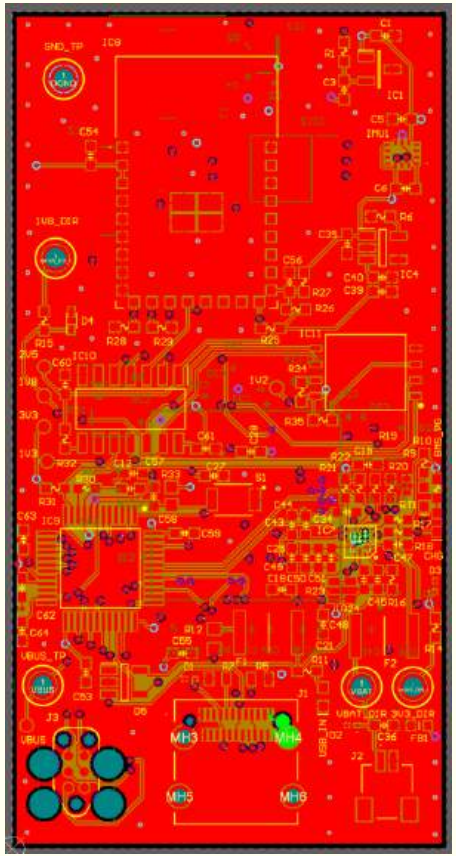
V0 Motion Tracking Collar PCB

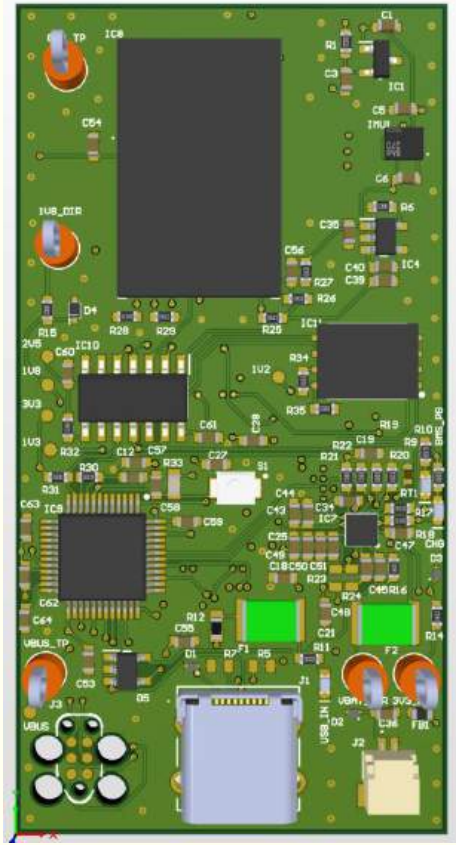


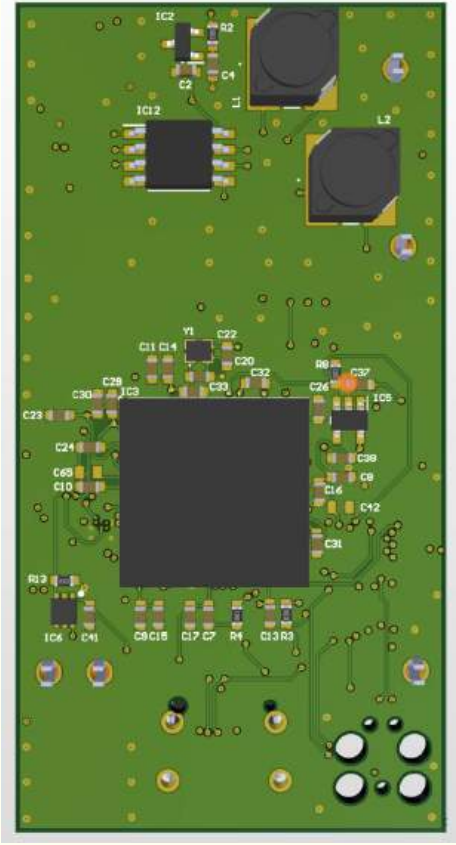


V1 Radar Heart Rate Collar PCB

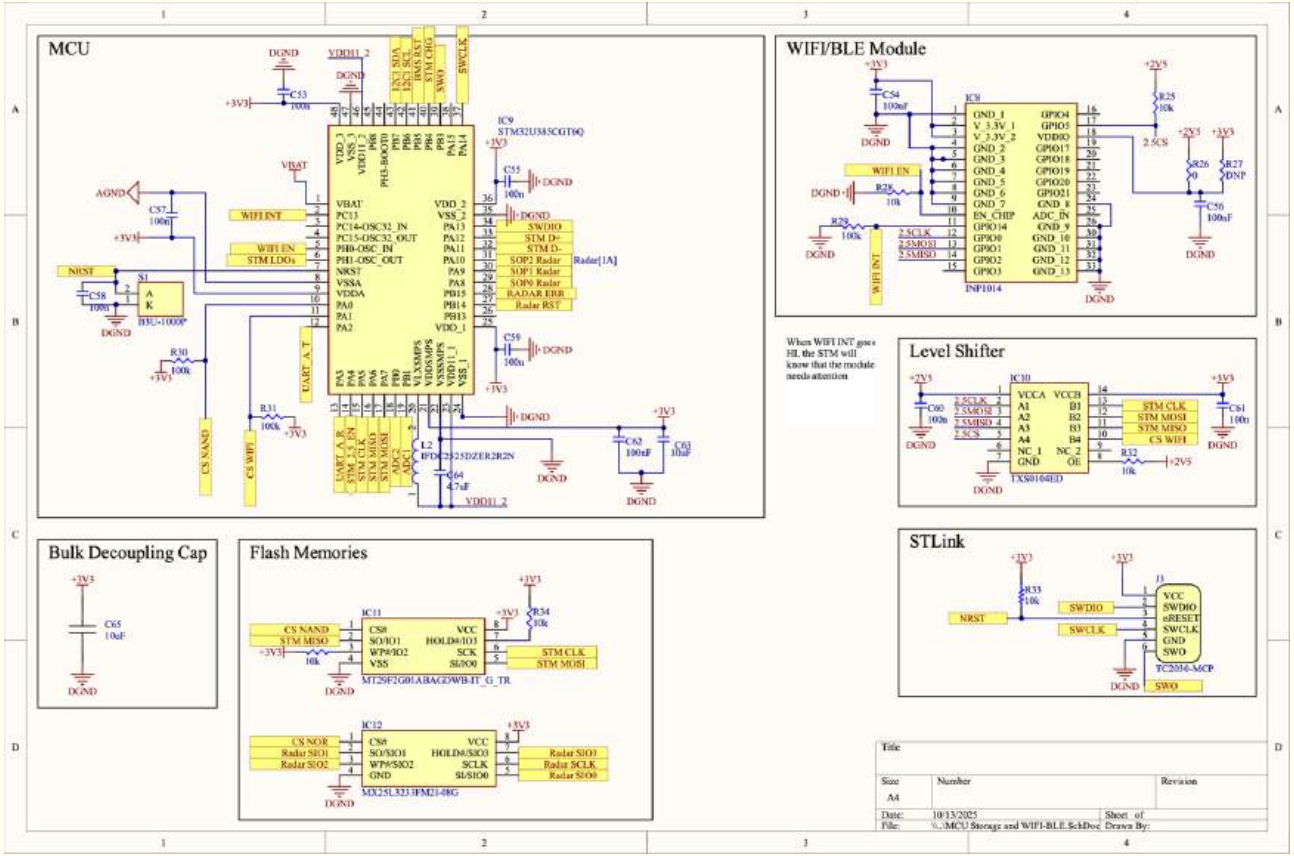




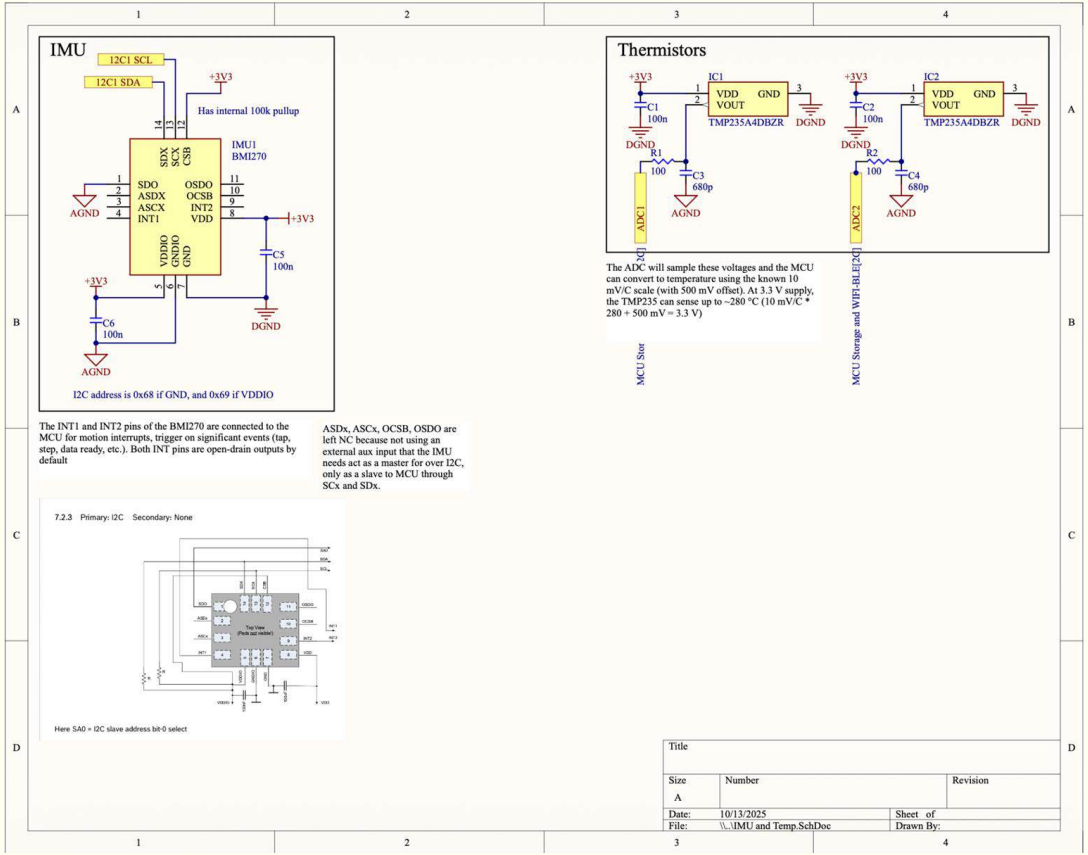




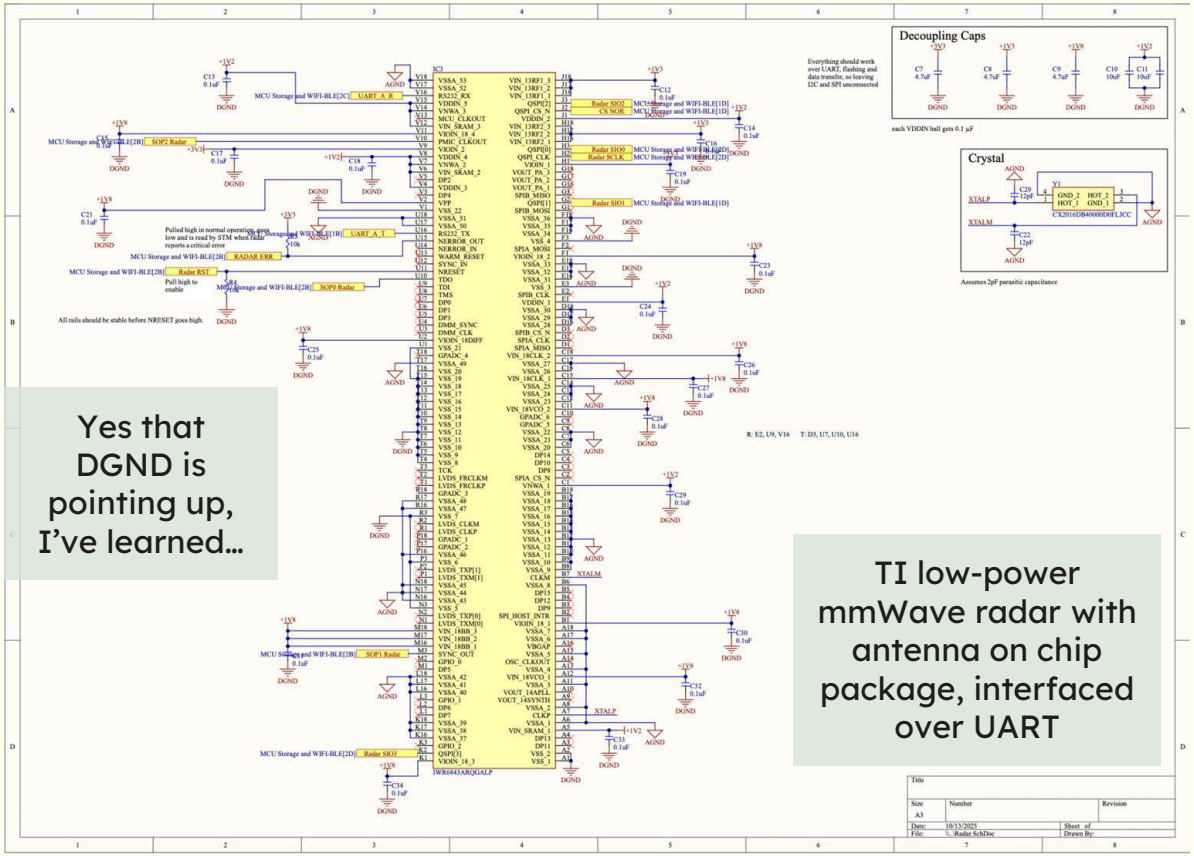
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